Doubt Class with Lakshay

Special class

Love Babbar • Sept 21, 2024

Maps and Tries Class - 1

Special class

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⇒ HashMaps → STL

HashMaps → D.S → <K, V>

Unordered map

map

hash[256]

string, double

Key	Value
Scorpio	7.3
Fortuner	11.2
Maruti	23.4
Creta	18.3

<String, double>

Map →

Map → pair → K, V

3 → "c"

3 → c

Map

Balanced BST

Tree -> array

travel -> sorted order

Ordered_Map -> O(log n)

Unordered_Map -> O(1)

array -> O(1)

Bucket Array

Print

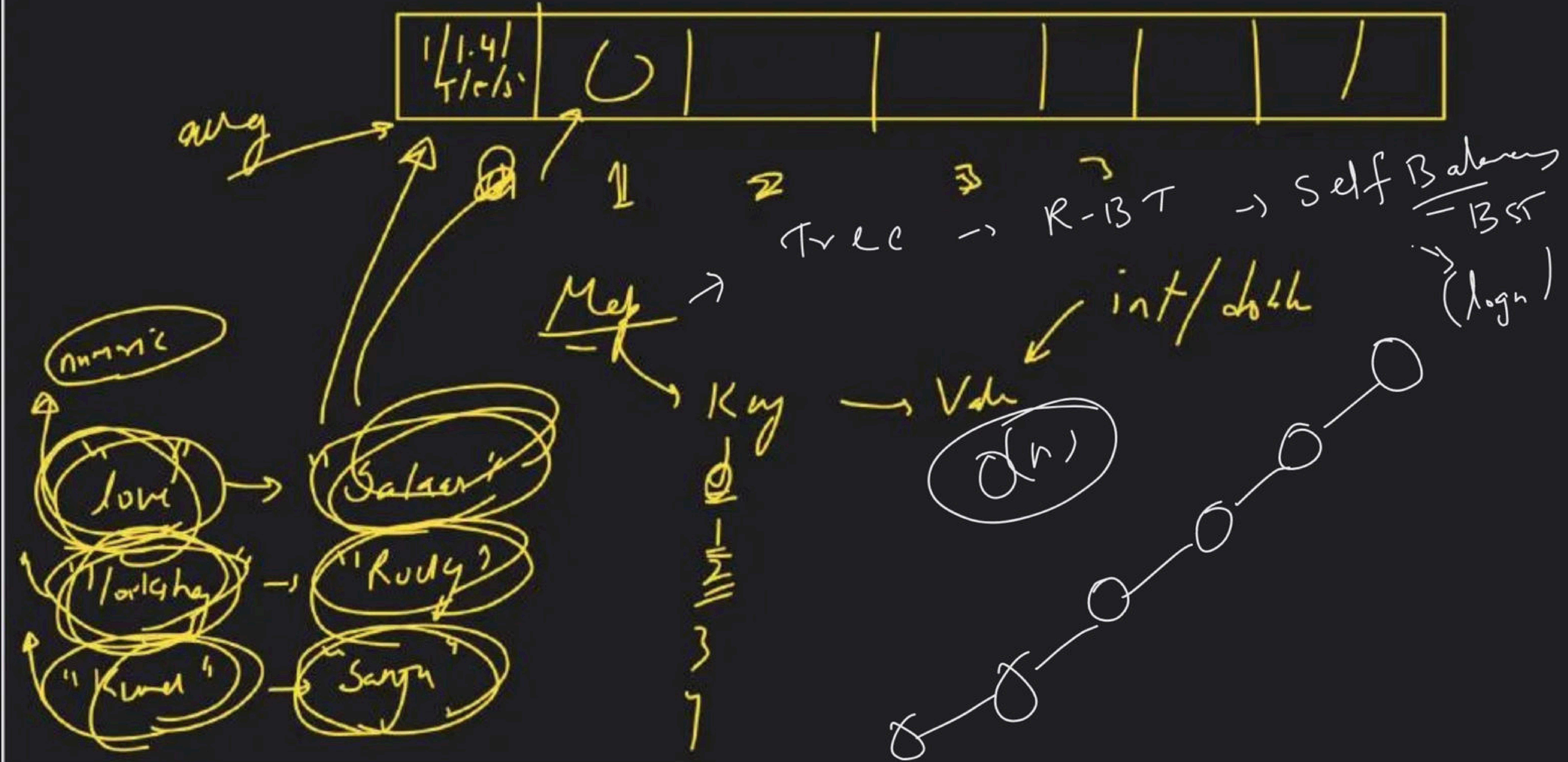
for(auto it : mp)

O.M

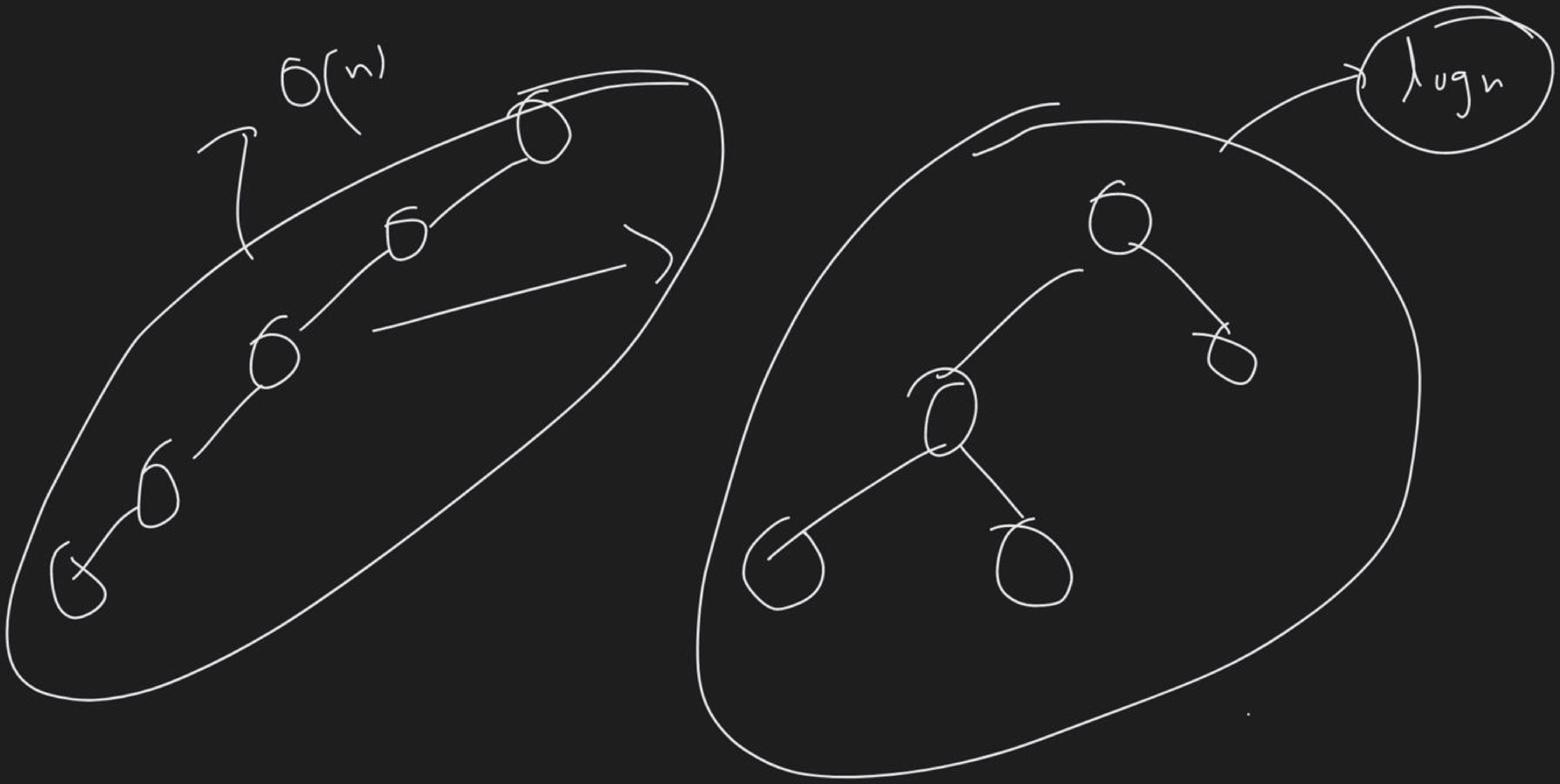
3 -> Scar

2 -> Fa

1 -> Betw



arr
Map
Key
Tree -> R-BT -> Self Balancing BST
(log n)
int/double
O(n)
numeric
love
Rudy
Sanju


O(n)
log n

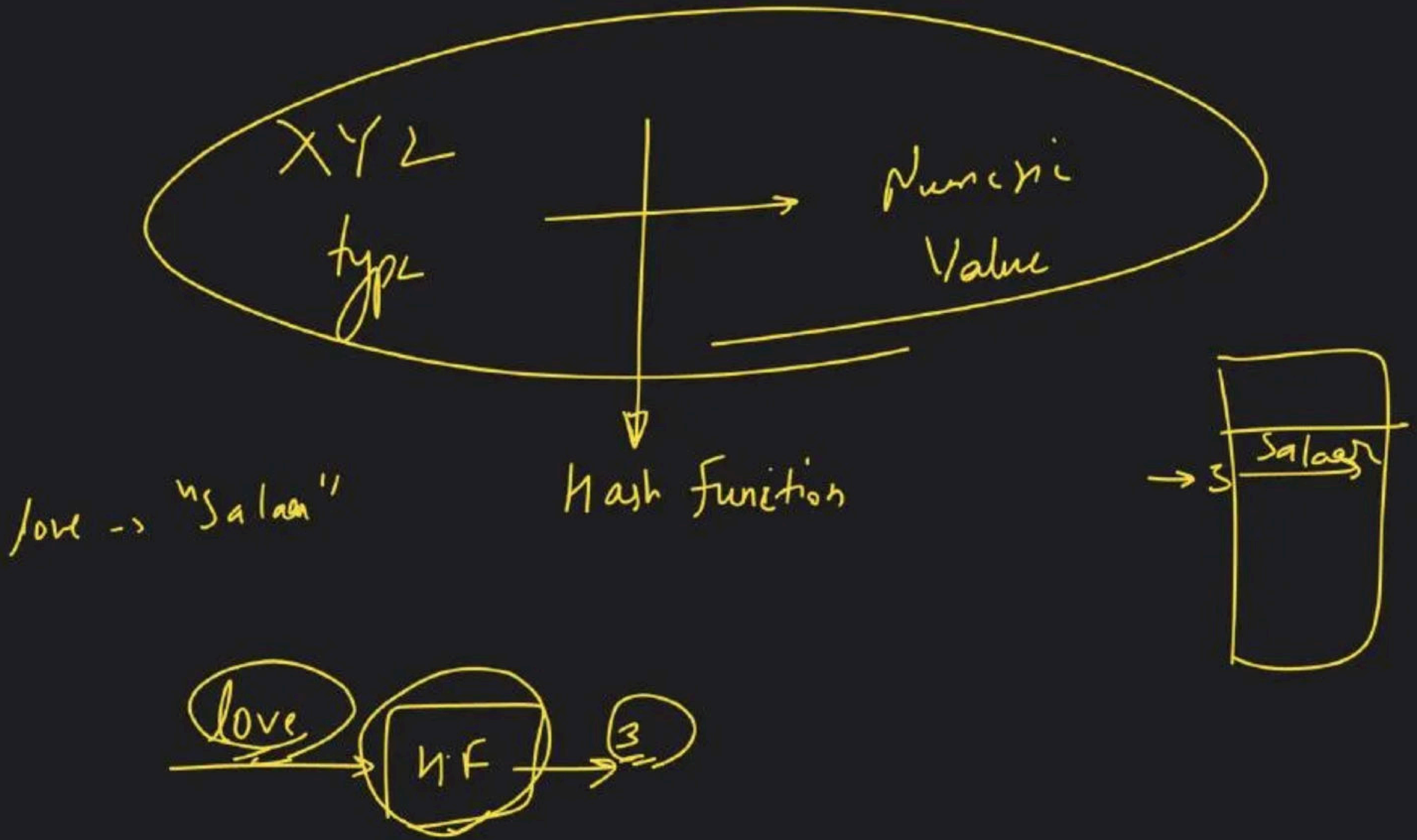
Scorpio

abcd $\rightarrow$ $(97 + 98 + 99)$

$= 294$ $\rightarrow$ Scorpio

cba $\rightarrow$ $99 + 98 + 97$ $\rightarrow$ Collision

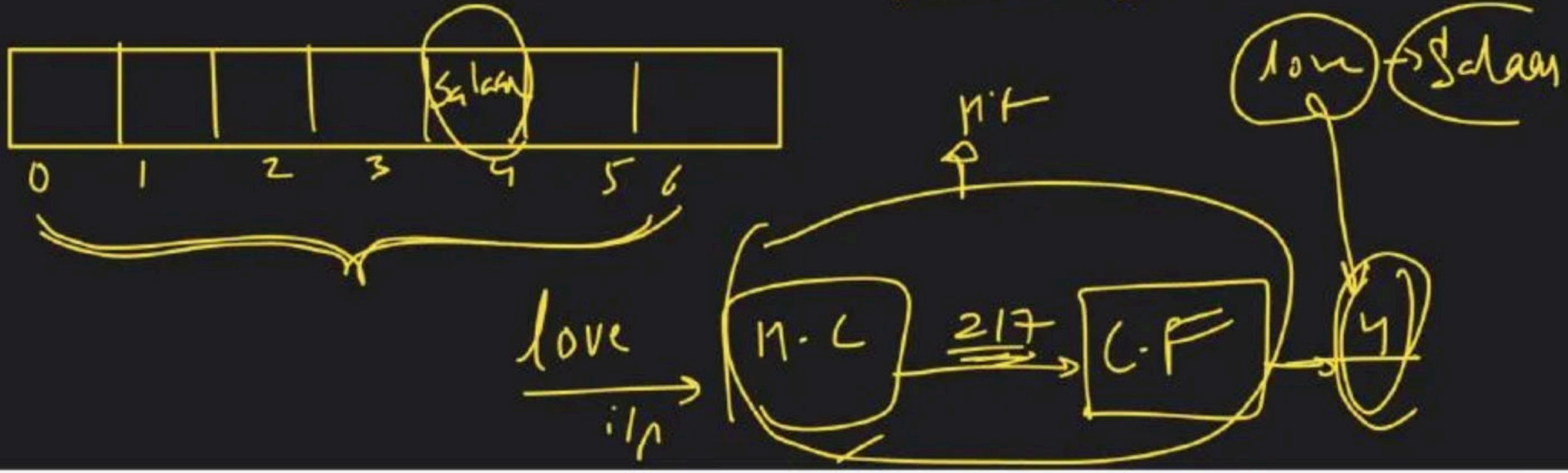
$= 294$



XYZ type
Numeric Value
Hash Function
love -> "Salaam"
love
HF
3
3
Salaam

Hash function

- Hash Code
- Compression function



Maps & Tries Class - 2

Special class

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→ Hash function

unordered_map

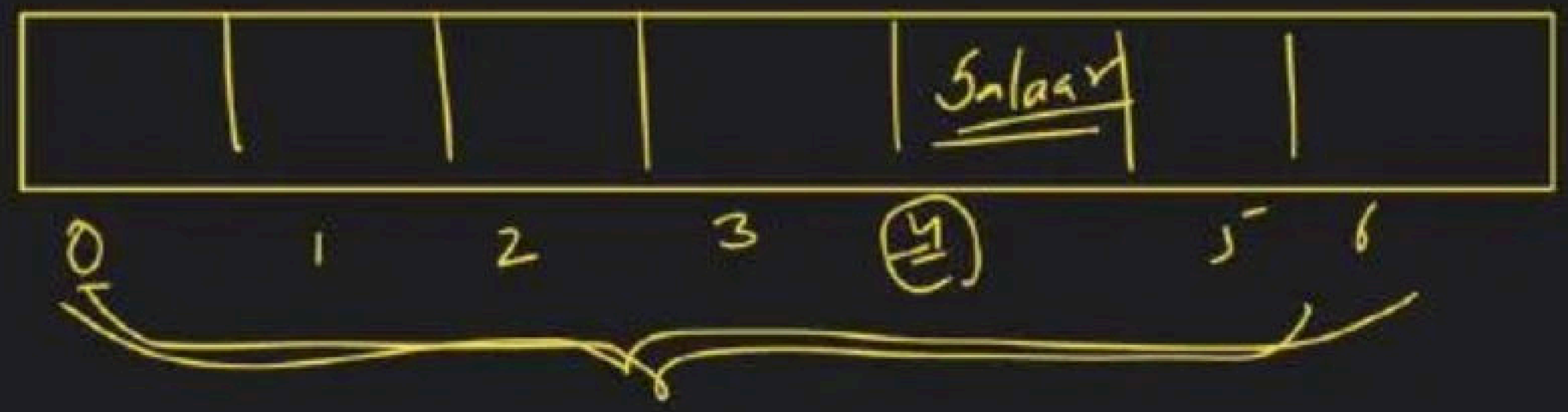
↳ Bucket Array

2 days → new link ↳ join

"love" → "Salaar"

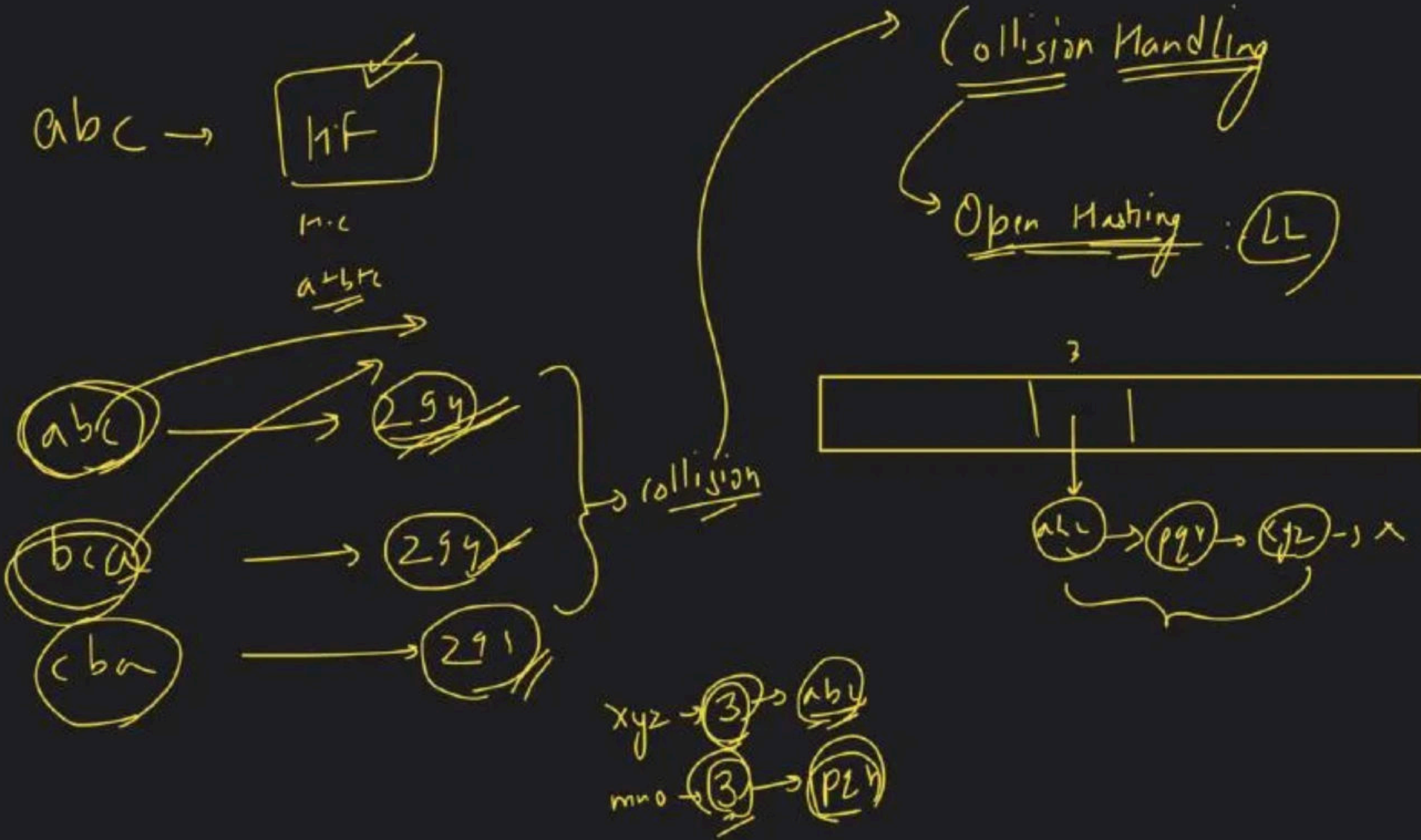
"Manish" → "KGF"

"Kunal" → "Rocky"



"love" → H.F → 4

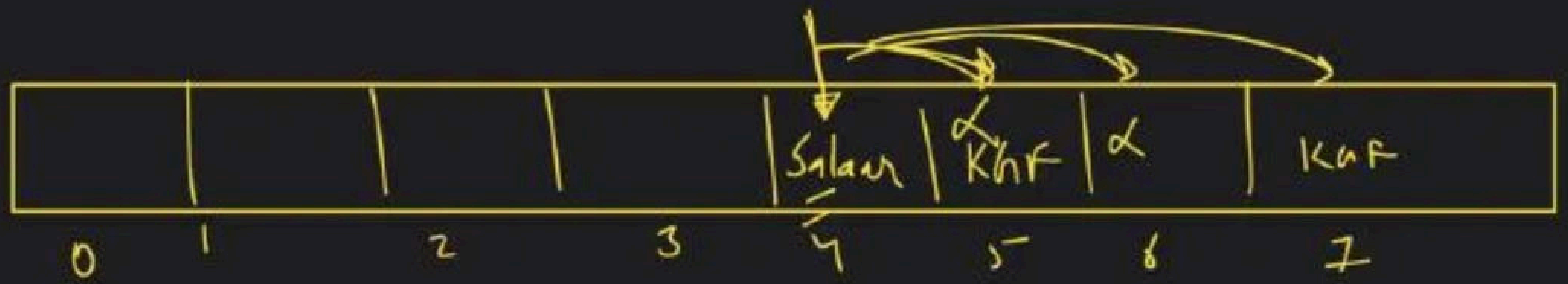
"love" → H.C → 241 C.F → 4


abc → HF
n·c
a+b+c
abc → 294
bca → 294
cba → 291
collision
Collision Handling
Open Hashing : LL
3
abc → pqr → xyz → ^
xyz → 3 → aby
mno → 3 → pzy

↳ Closed Addressing :→ [Next free space]

cubic Probing

$h(i) + i^3$



Linear Probing

love $\xrightarrow{4}$ Salaar

Manish $\xrightarrow{4}$ KGF

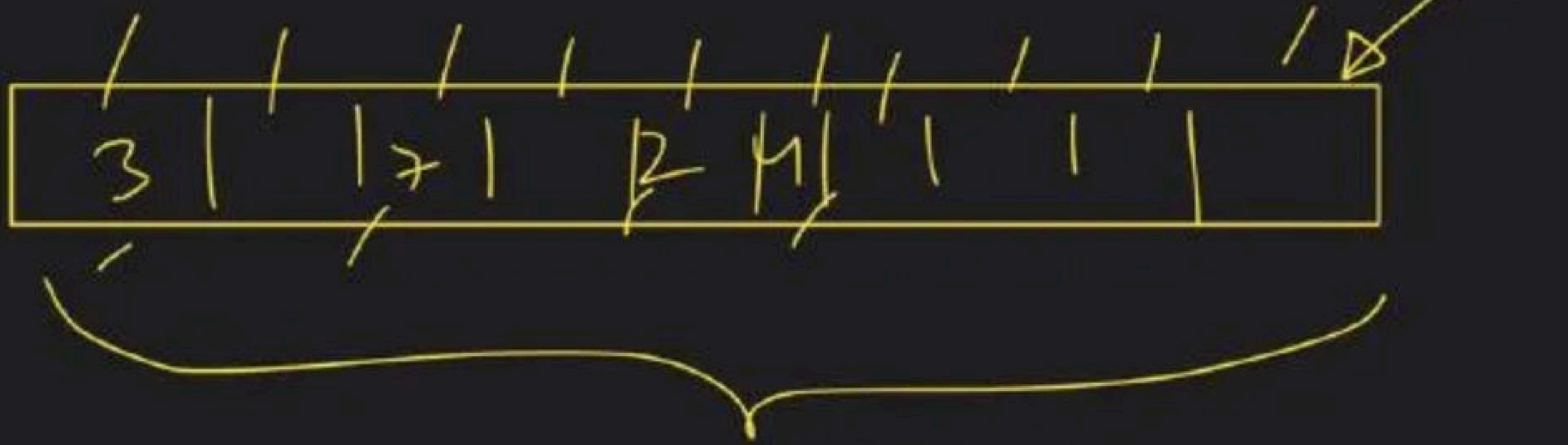
$h(i) + i$

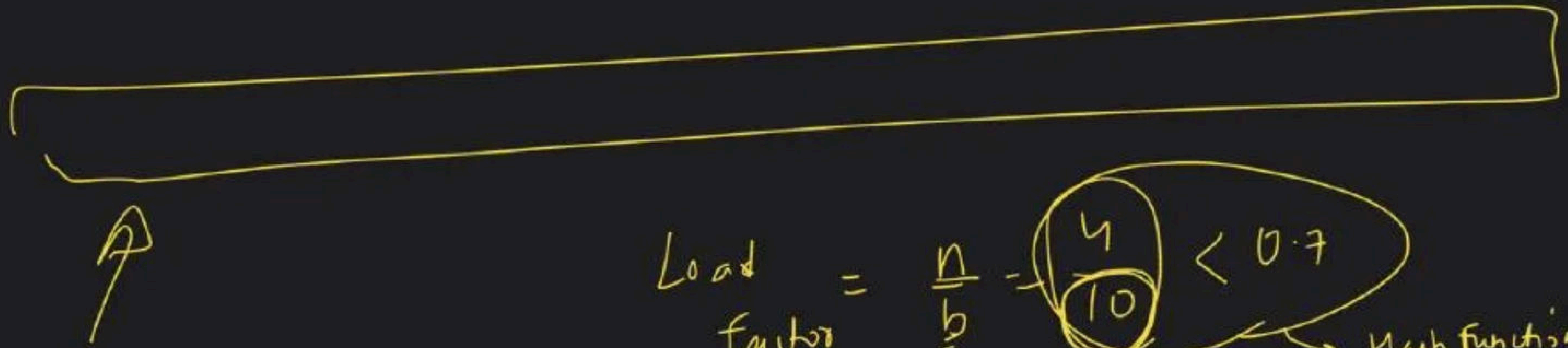
$i \to 1, 2, 3, 4, 5$ - - .

$h(i) + i^2$

Quadratic Probing

↳ $i \to 1, 2, 3, 4$ - - -



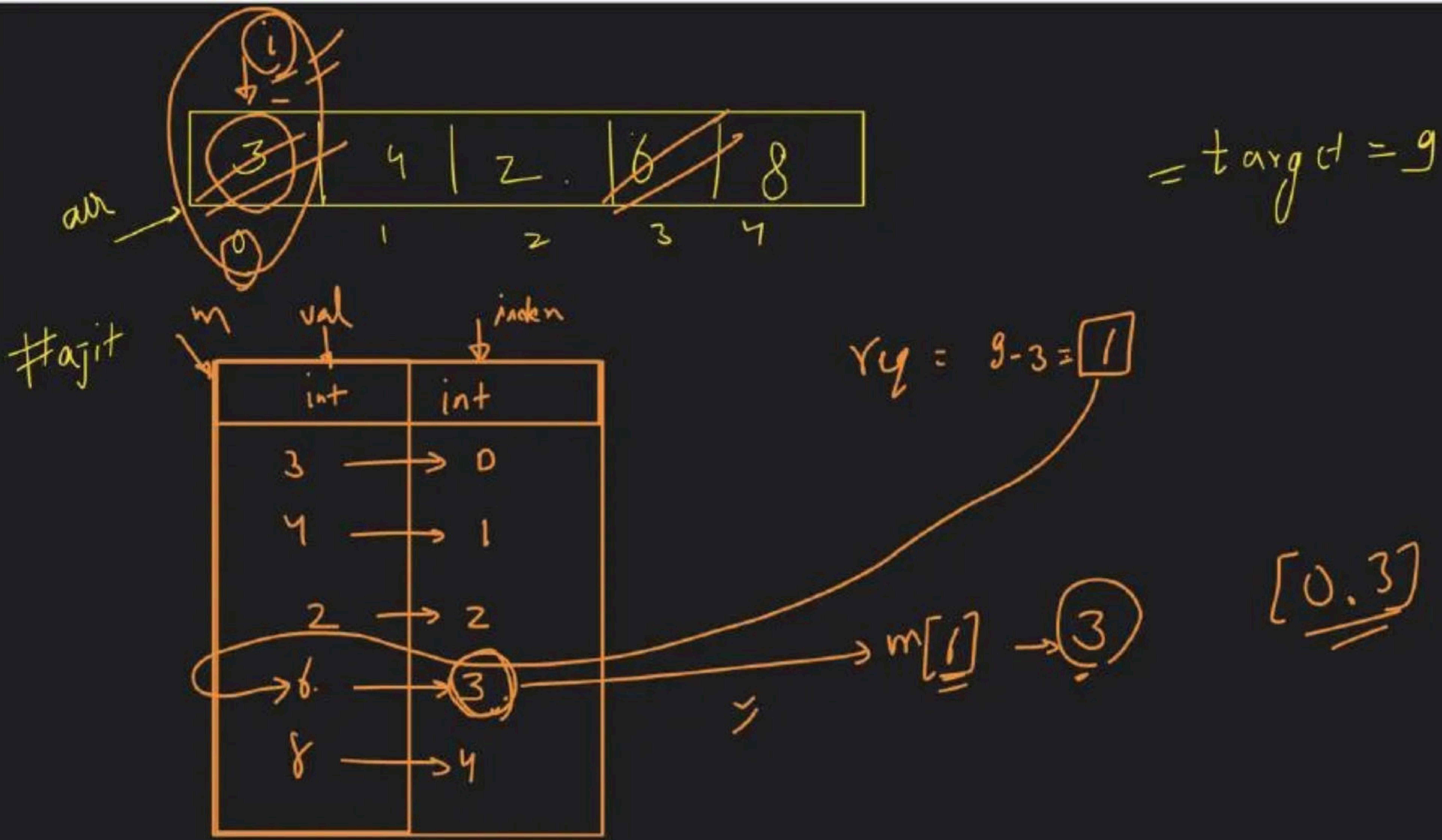


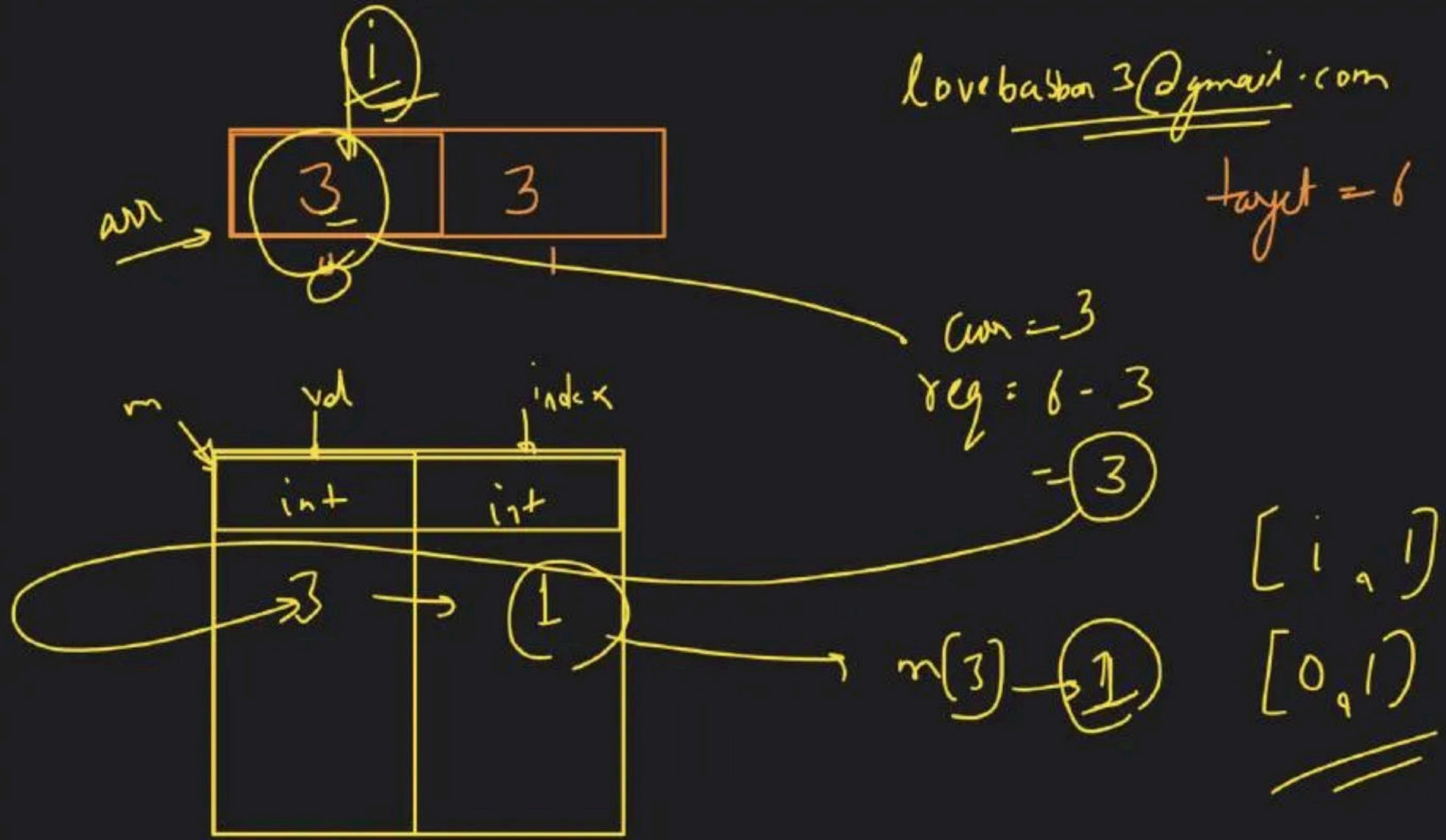
Load factor $= \dfrac{n}{b} = \dfrac{4}{10} < 0.7$ → Hash Function currElement + x = target

x = target - currElement

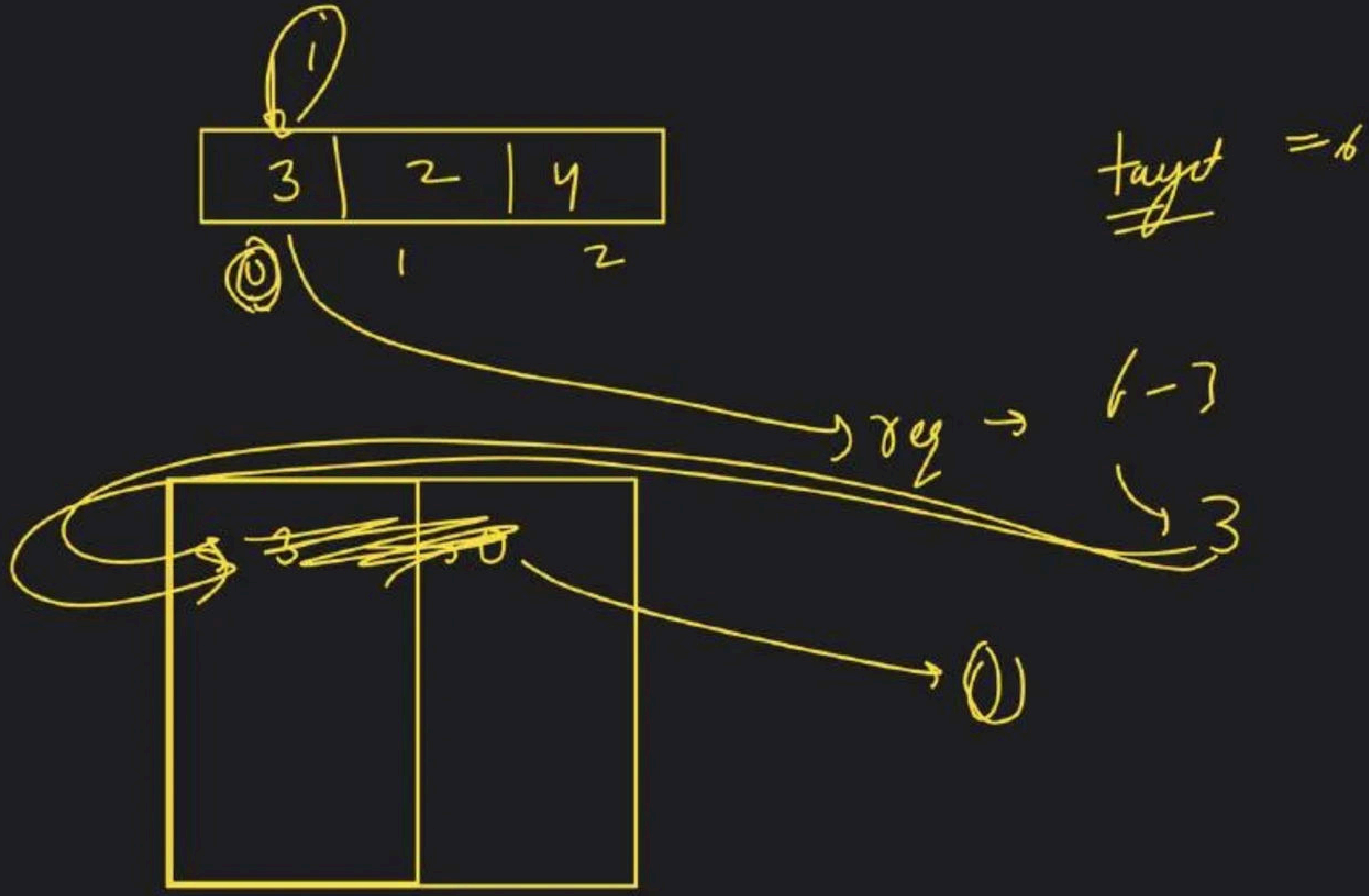
value

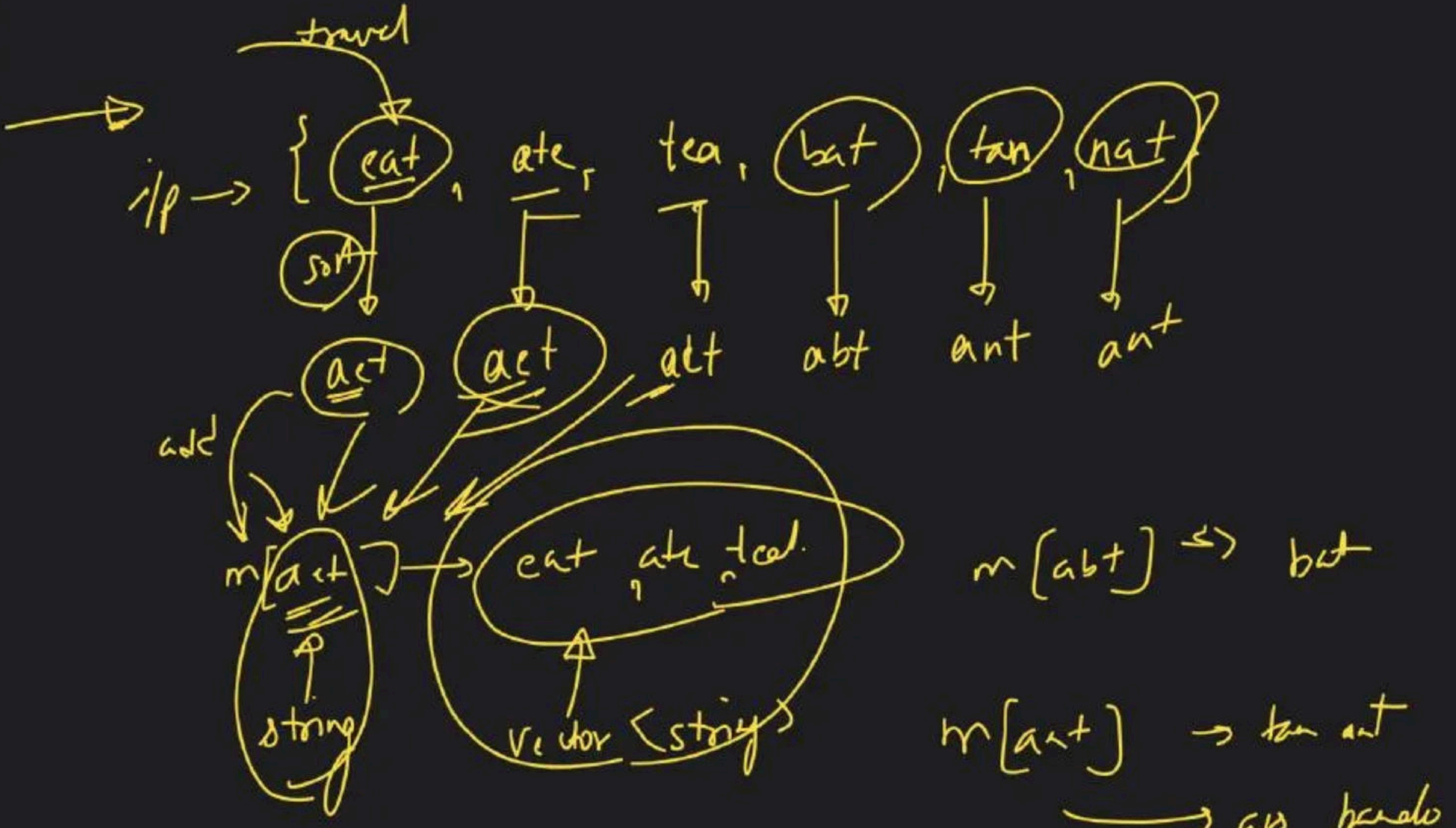
map - ans


i
arr
3 4 2 6 8
0 1 2 3 4
target = 9
#ajit
m
val
index
int
int
3 → 0
4 → 1
2 → 2
6 → 3
8 → 4
req = 9-3 = 1
m[1] → 3
[0, 3]



i
arr
3
3
lovebabbar3@gmail.com
target = 6
curr = 3
req = 6 - 3
= 3
m
val
index
int
int
3
1
m[3] → 1
[1, 1]
[0, 1]


1
3 | 2 | 4
0 1 2
target = 6
req → 6-3
3
0



travel
i/p →
{eat, ate, tea, bat, tan, nat}
sort
act
act
act
abt
ant
ant
add
m[act] → eat, ate, tea
string
vector<string>
m[abt] → bat
m[ant] → tan nat
ans banalo

{ tan, ate, bat, eat, nat, tea }

original = ~~tan~~ "" a~~te~~ bat eat nat tea

copy = ~~tea~~ ate bat eat nat tea

sort sort sort sort sort sort

copy = ant act abt act ant act

ans → {tan, nat}

{ate, eat, tea}

{bat}

string	vector<string>
ant	tan, nat
act	ate, eat, tea
abt	bat

i/p →

s = " bar foo the foo bar man "

words = ["foo" , "bar"]

→ guarantee

↳ Concatenate String

2+3 → $\boxed{6}$

start

bar foo the foo bar man

b arf oot hef oob arm an

ba rfo oth efo oba rma n

word[0].length

start < word[0].length

offset → 0, 1, 2

① map → track word count

word → 2
good → 1
best → 1

words[] → word good best word

→ remove

word good good good best word

② Break string

visited

word → 1
good → 1
best → 1

count ++

① map → track word count
② Break string

Kya ab mila

→ substring → validate
include
extra
remove

Count = words.size()

abs nhi

2 > 1 → removed

map
↳ good → 1

str → substr(i - ((cnt - 1) * wordLen))

word good good good but word

vis
↳ word → 0
↳ good → ~~X~~ ~~2~~ 1

i

word good good

cnt -- wordLen = 4
cnt --
count =

0 1 2 3 4 5 6 7 8 —

words → foo, bar

wordLen = 3
count = 2

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14

foo bar the foo bar

i

VM
foo → 1
bar → 1

count = 2

i − (count − 1) * wordLen

words → foo bar

map
↳ foo → 1
↳ bar → 1

bar foo the foo bar man

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

0

9

~~visited~~
~~↳ bar → 1~~
~~↳ foo → 1~~
~~count = 1~~ 2

~~visited~~
~~↳ foo → 1~~
~~↳ bar → 1~~
~~count = 2~~

visited → { }

count = 0

1 → invalid substring

2 → jada elements me
valid mil jati

Remove

Maps & Tries Class - 3

Special class

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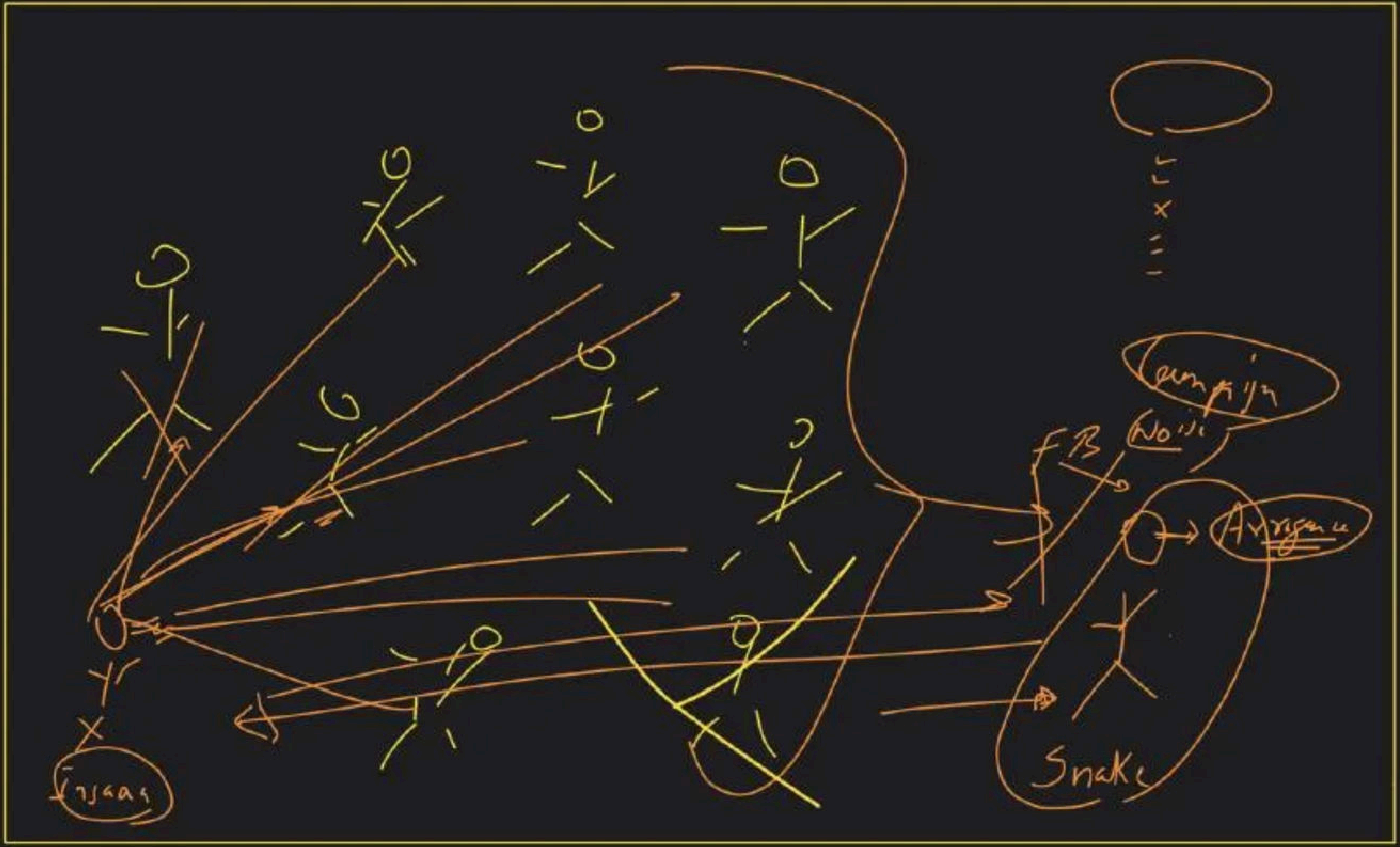
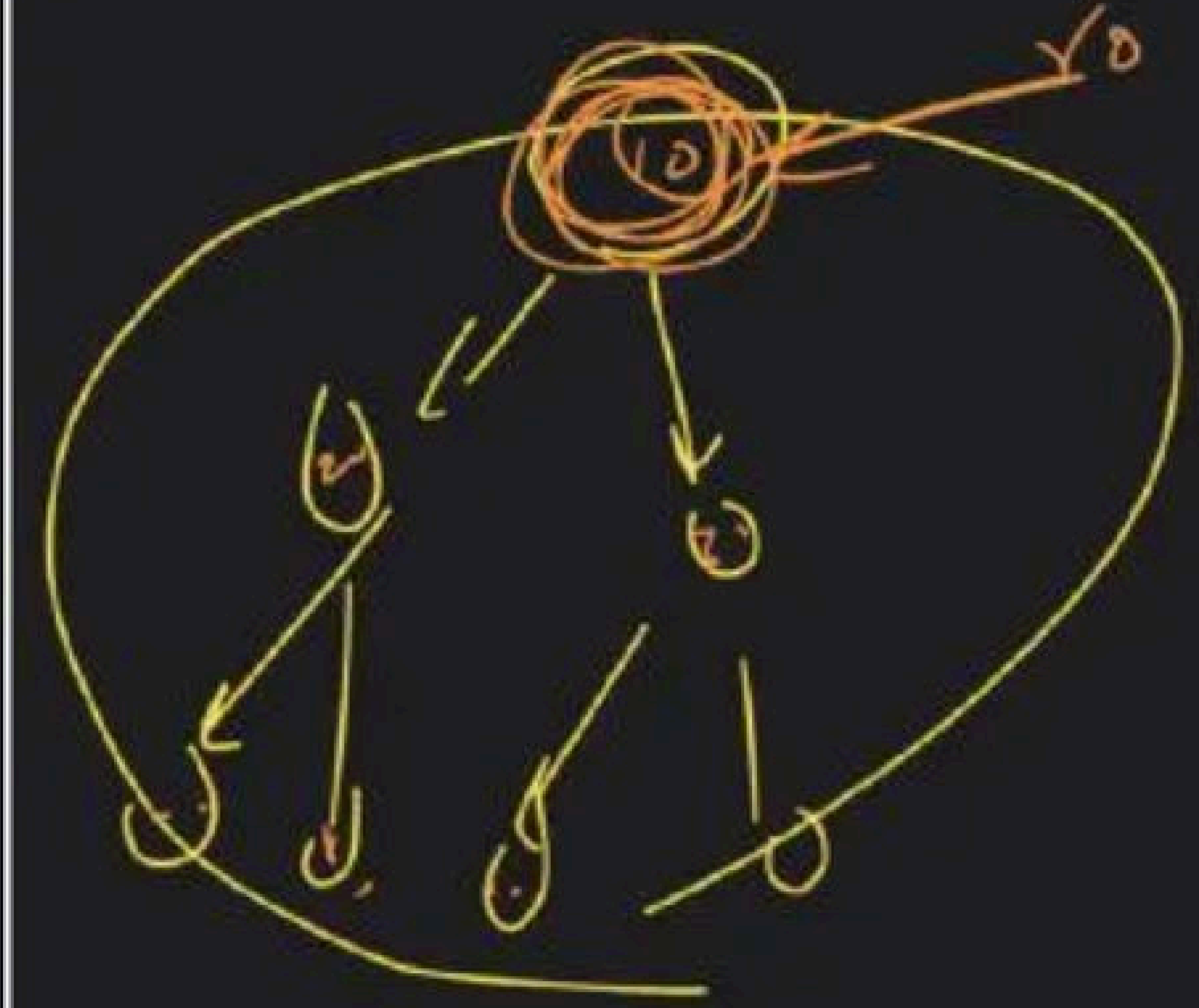

Snake

Tries:-

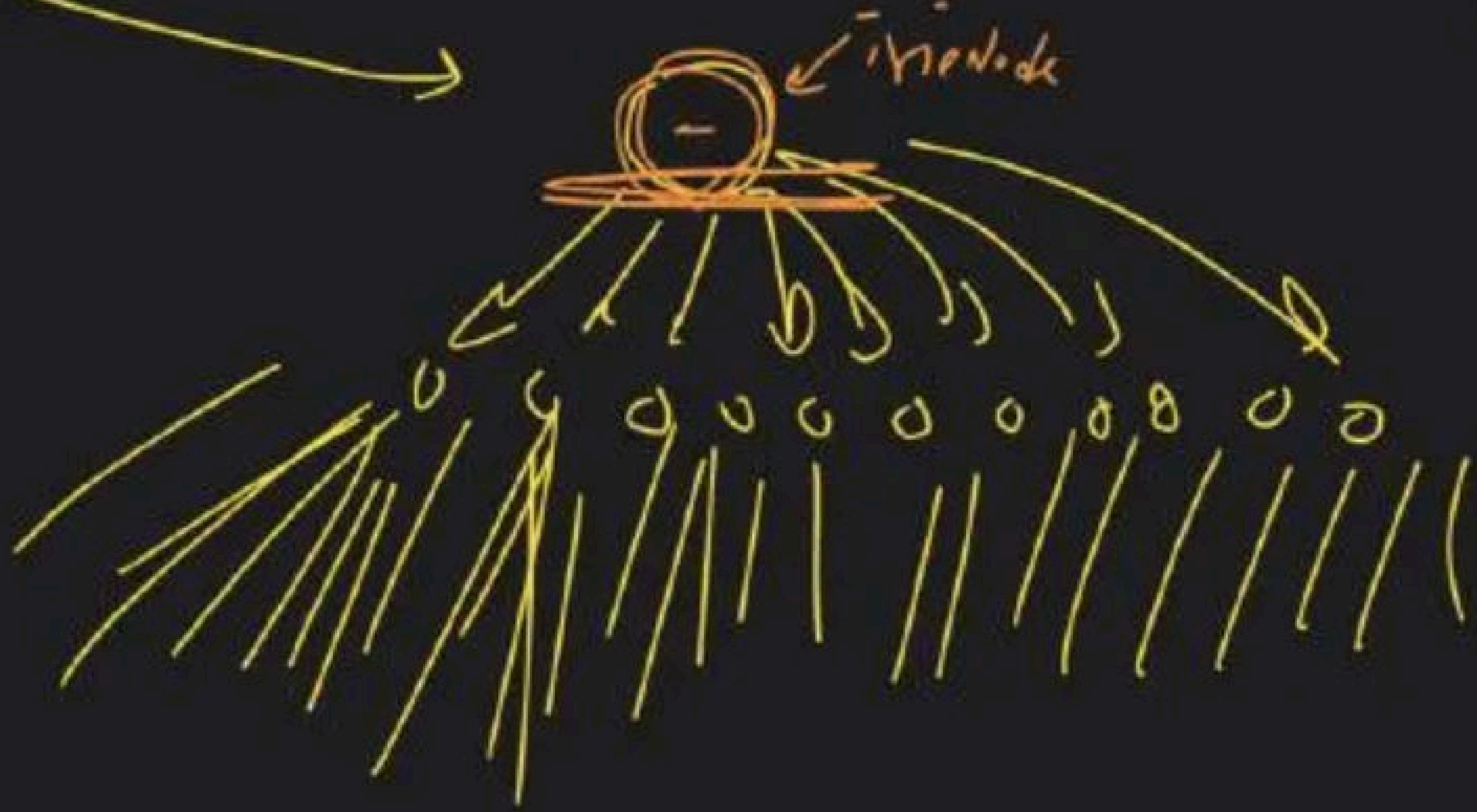
multiway-tree-DS

D.S

Pattern matching



Tree



Trie

insert

Search

remove

→ $O(L)$

→ $O(1)$

absent → create

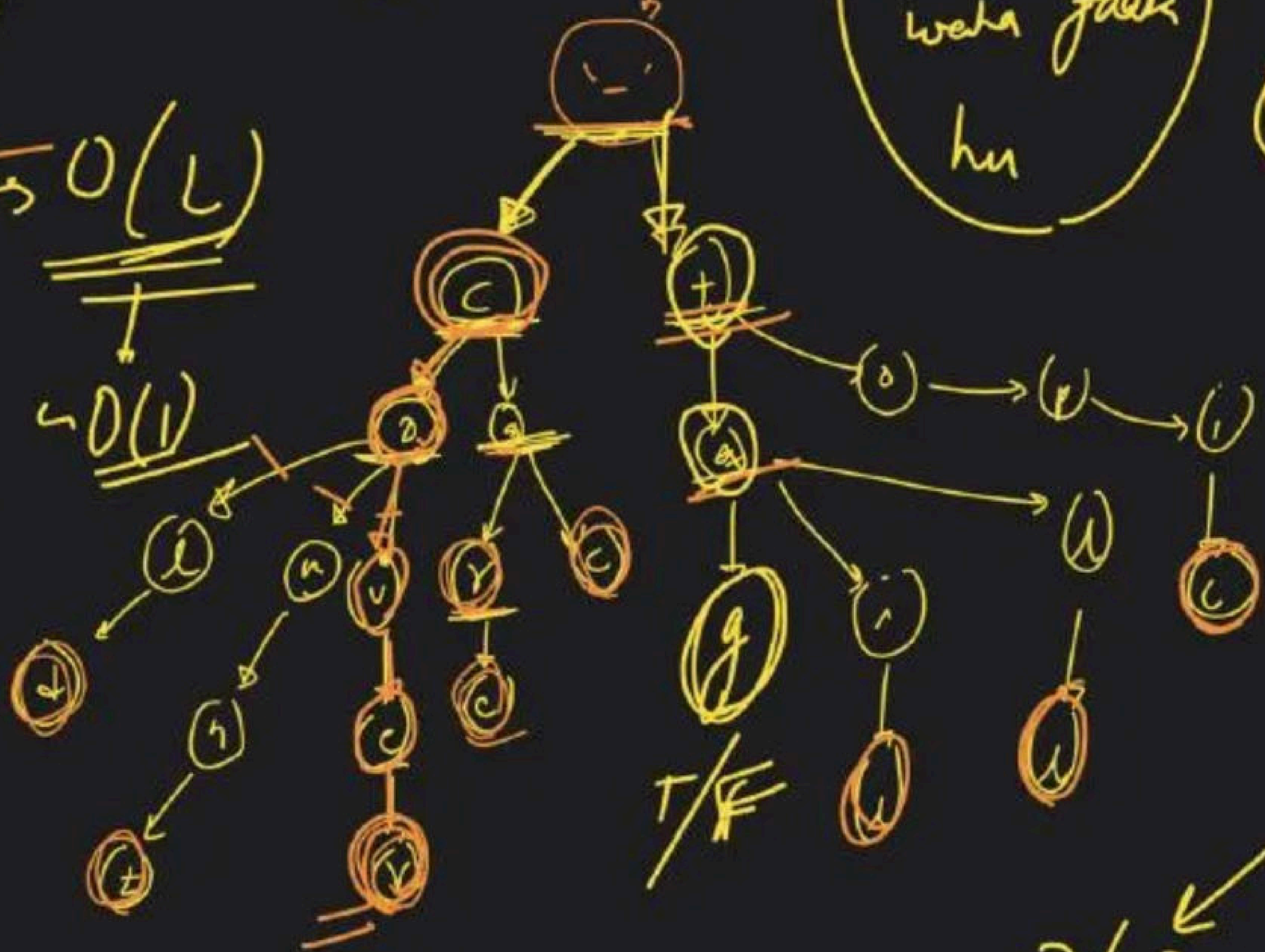
present → waha chale jao

1 case Baaki

Recursive

terminal

$n > L$



present → waha jaake hu

Terminal ↓ T→F

absent return

tag

g → terminal

delete

$O(L)$

Search

isTerm T→F

abc

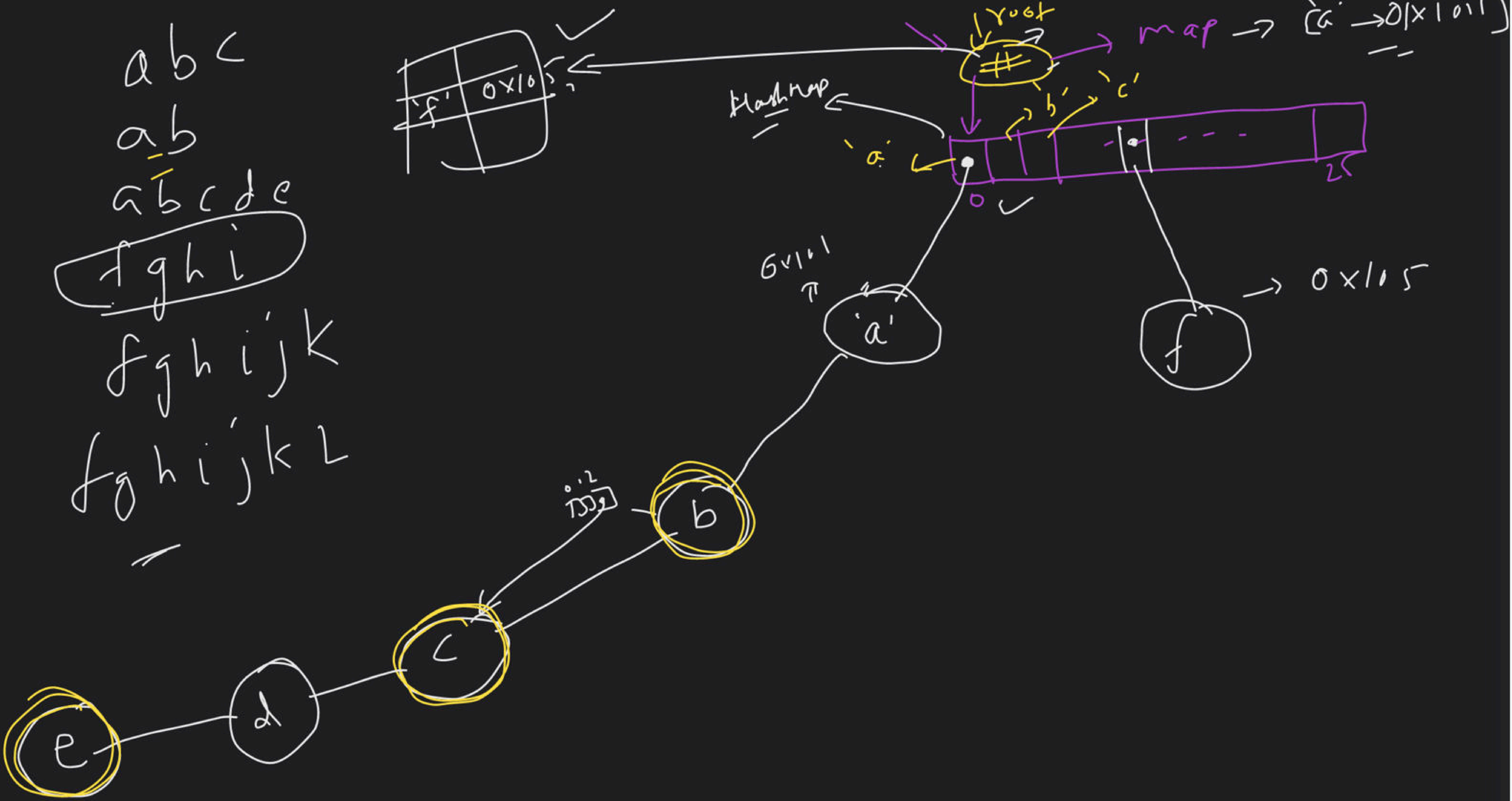
ab

abcde

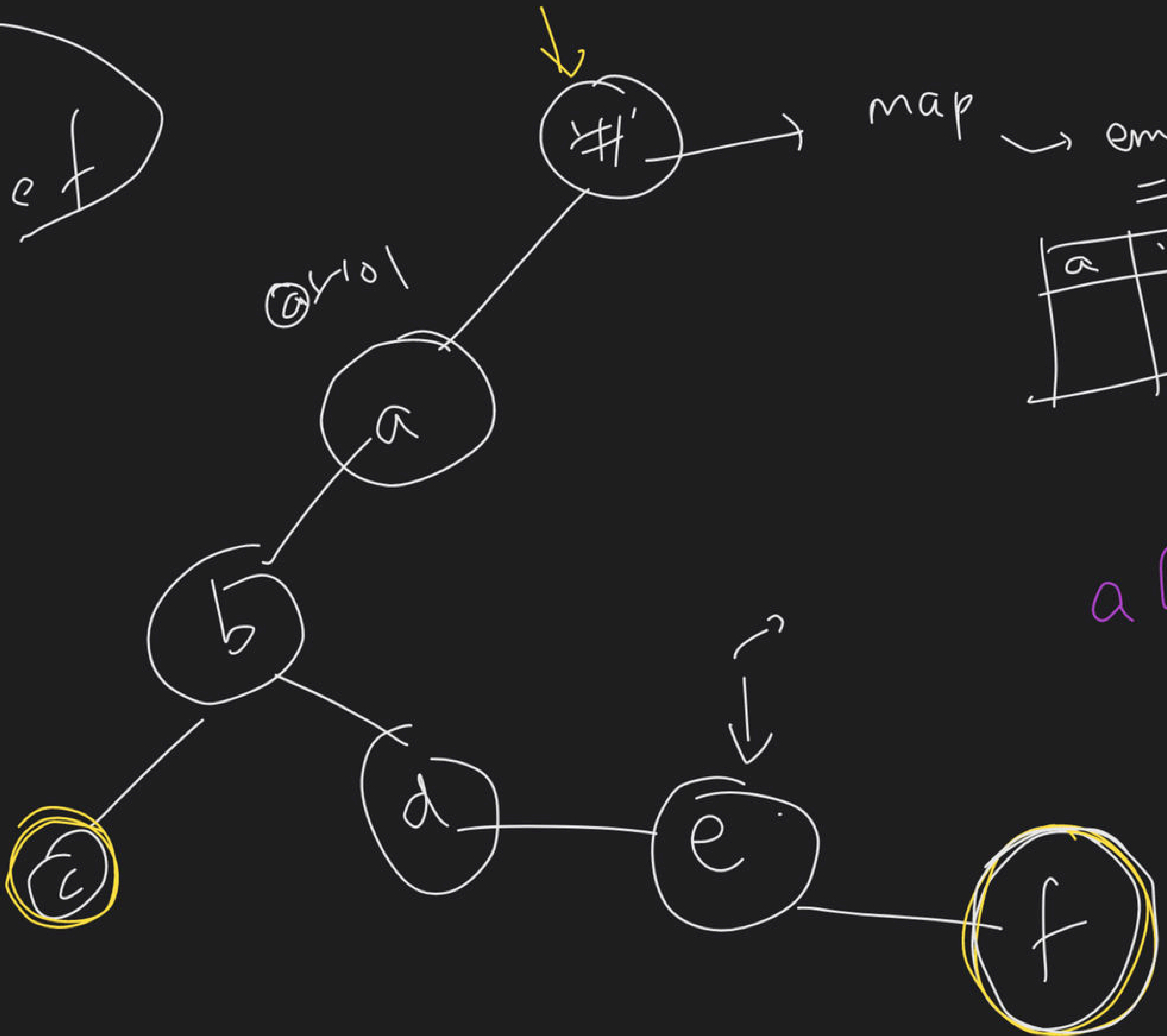
fghi

fghijk

fghijkl



abc
abdef



map → empty
=

a	'0xx'

abdef

→ a b c d e f g h i) new

→ b c d e f) new

→ c d e

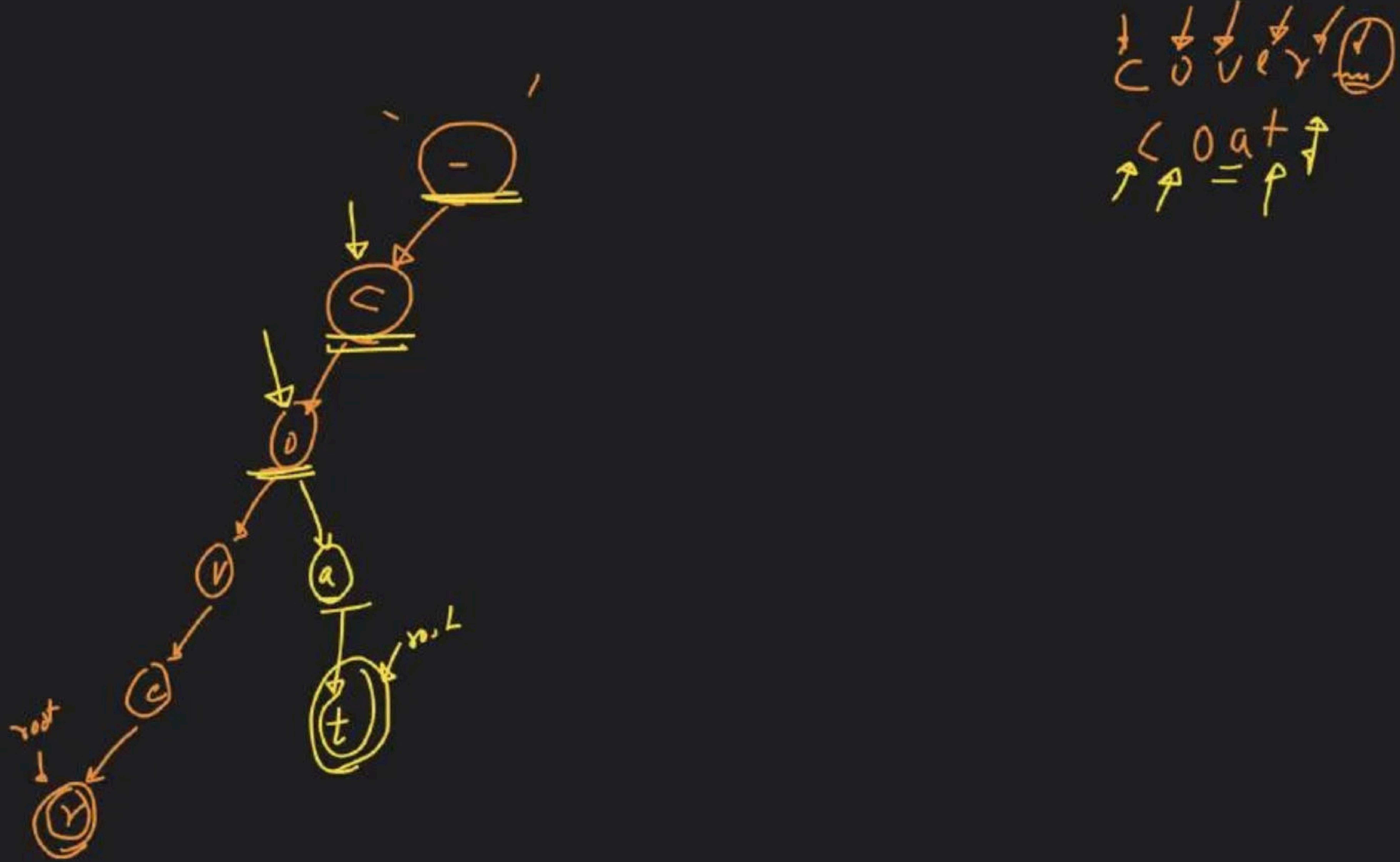
TrieNode

→ char data

ASCII value　　　address

→ children → map <int, TrieNode*>

→ bool isTerminal


cover
coat
root

Maps & Tries Class - 4

Special class

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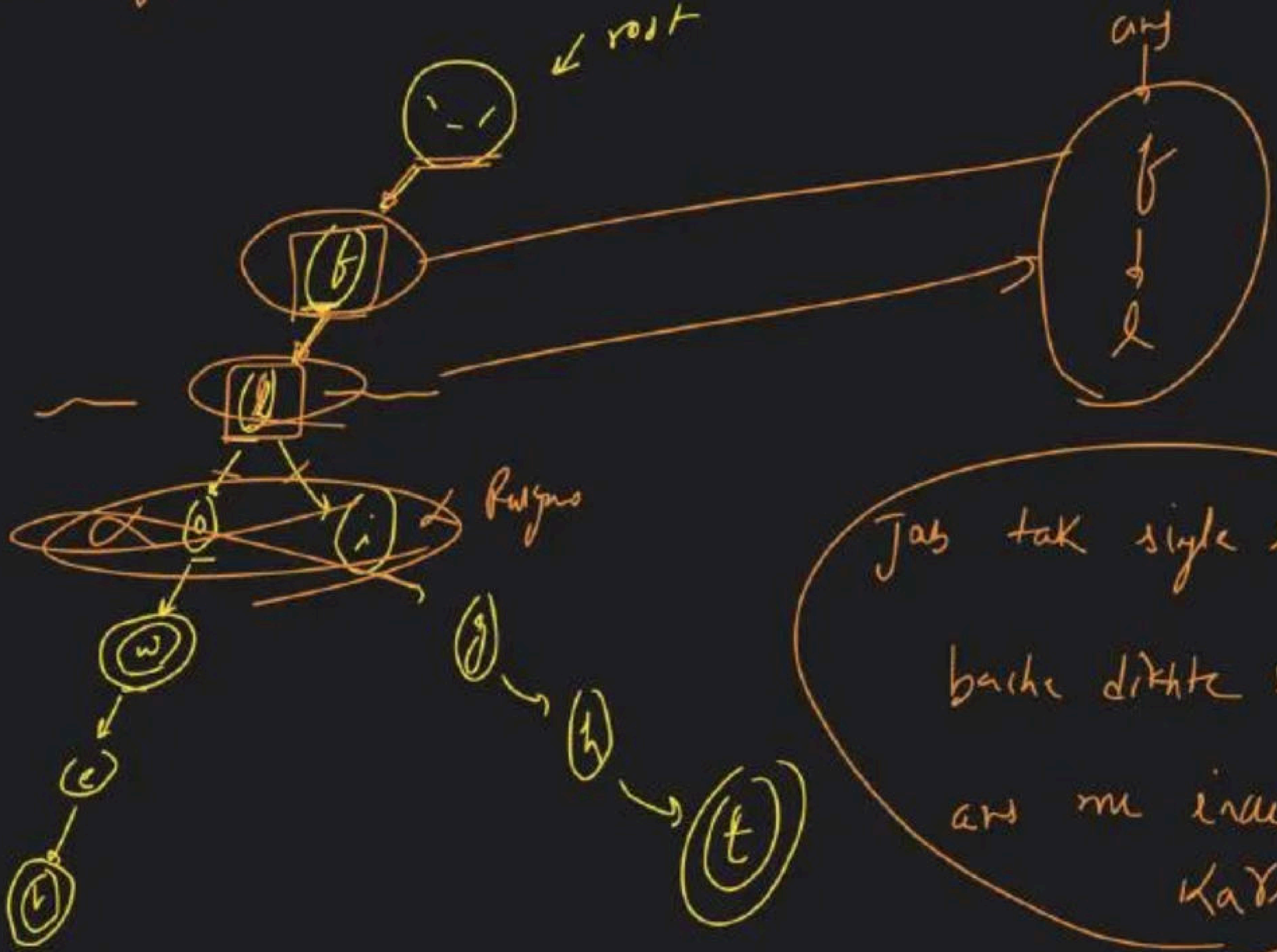
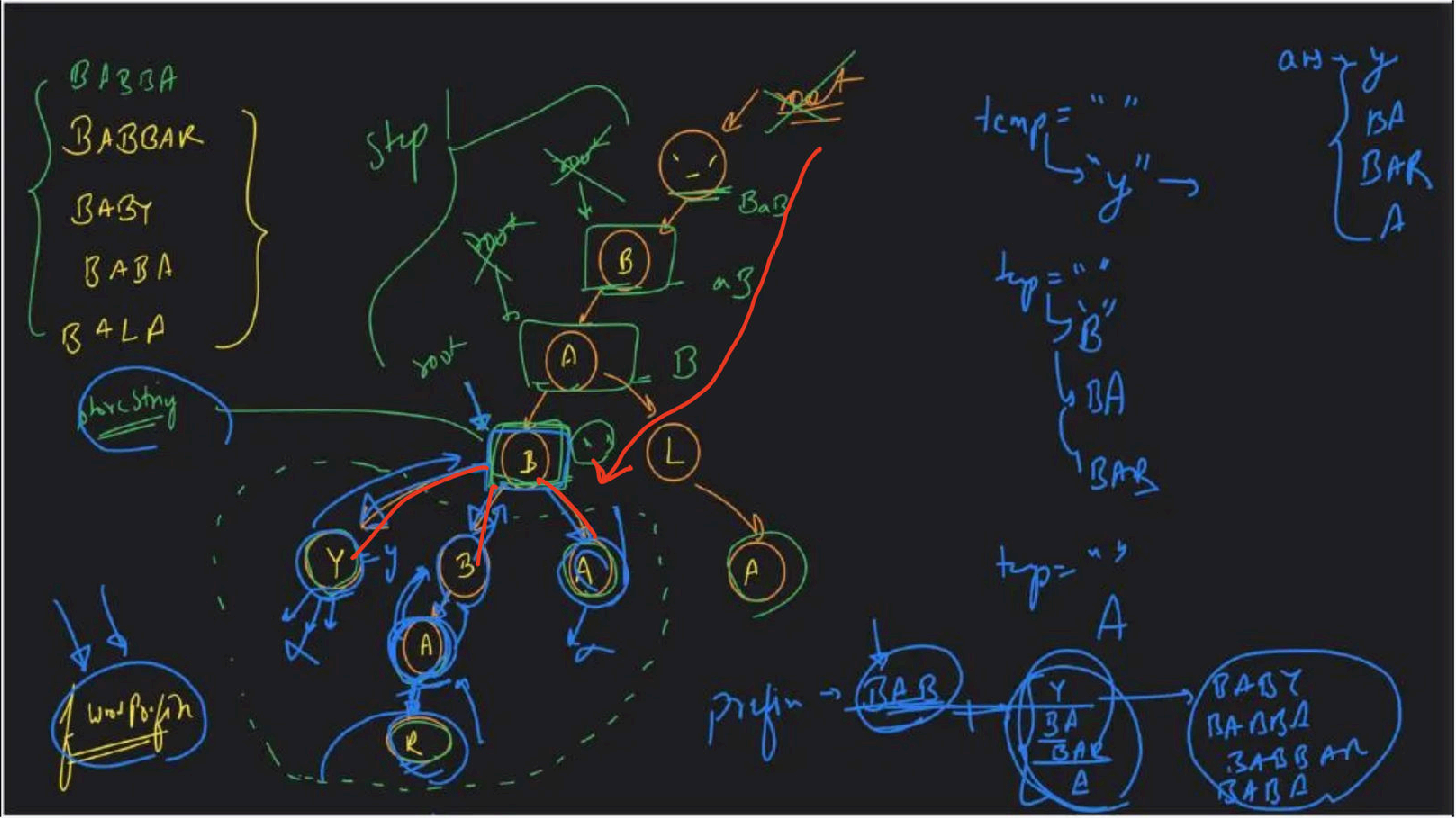
→ ~~flower / flow / flight~~

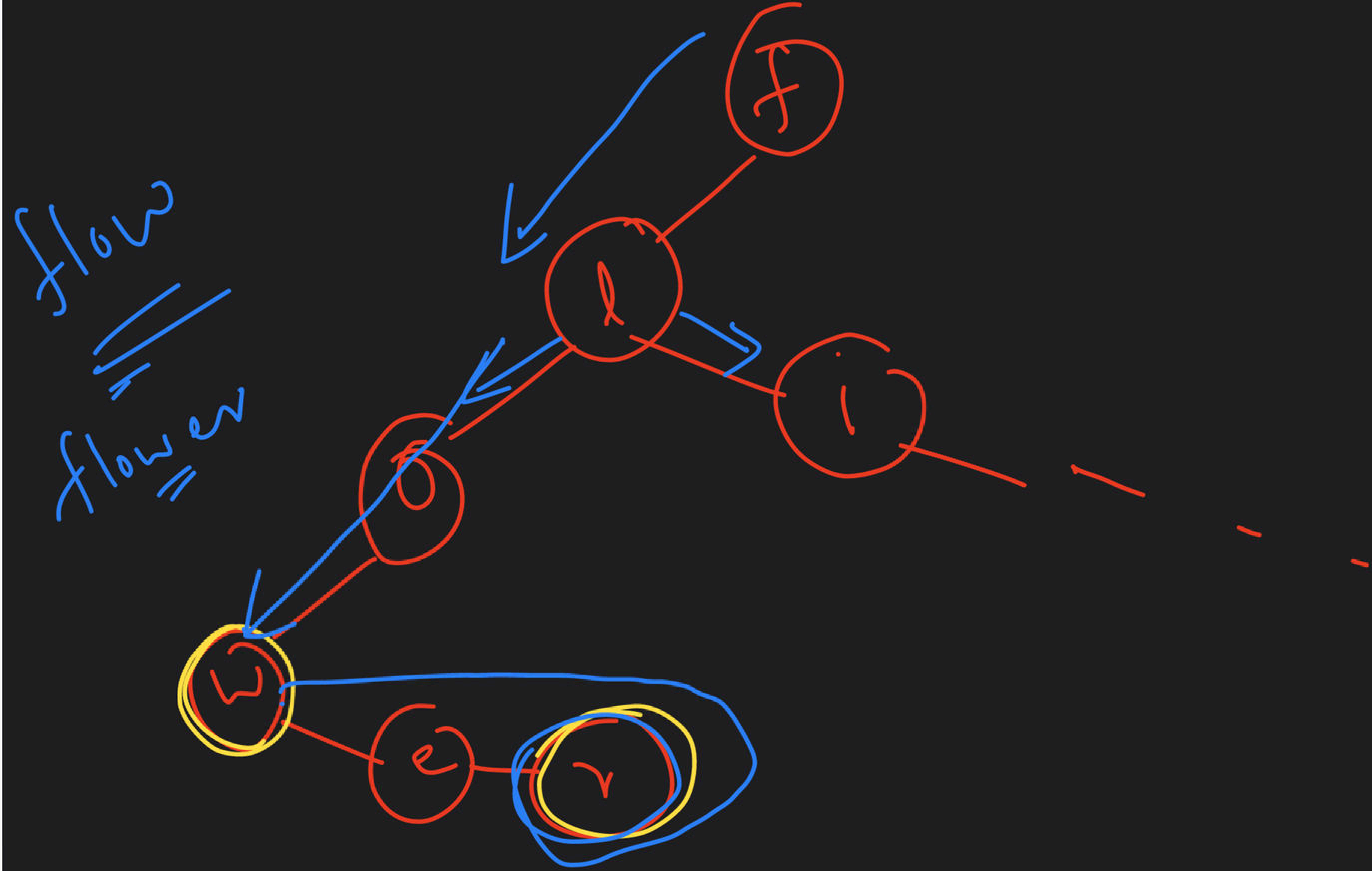


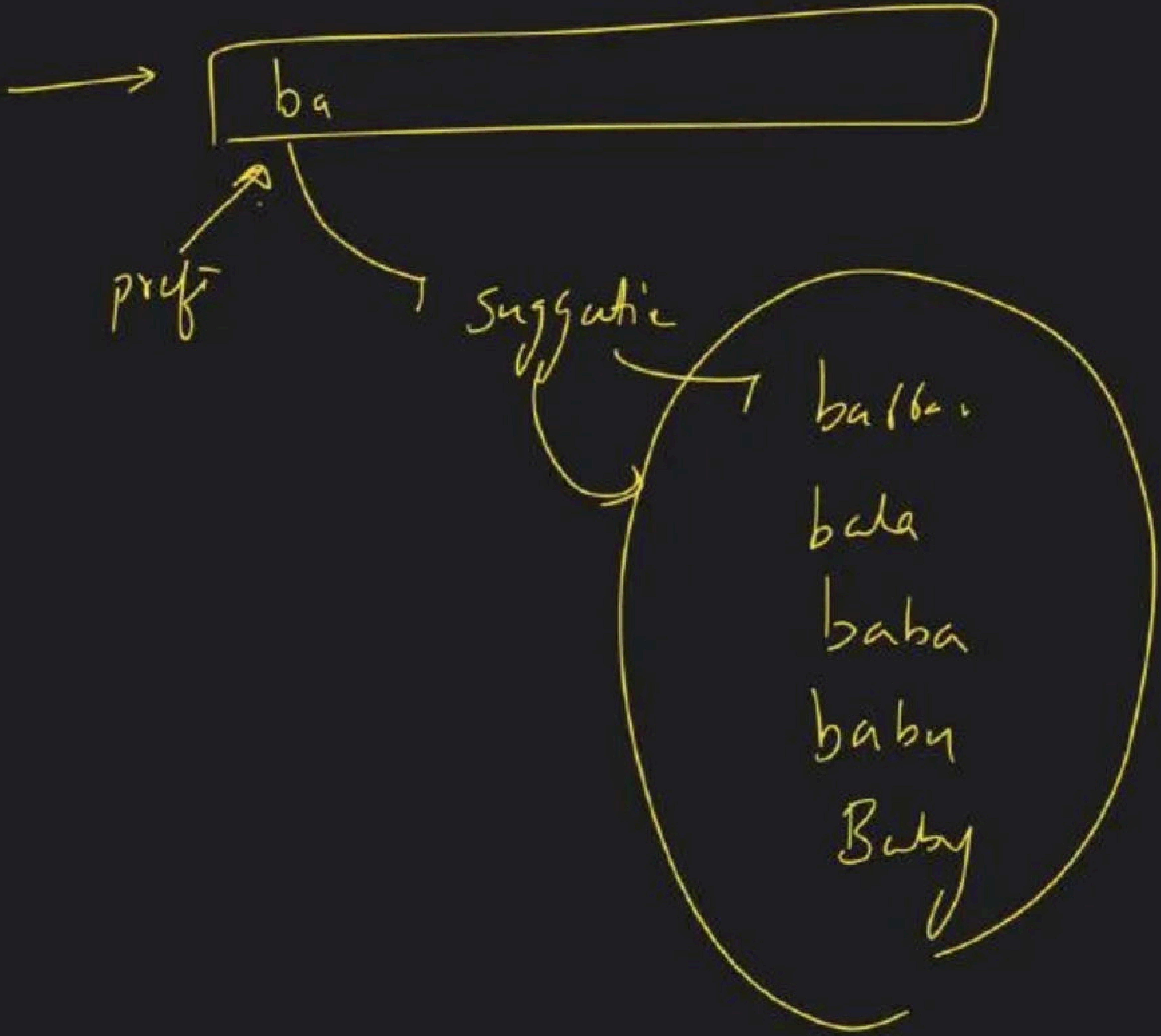
Jab tak single single bache dikhte rahe, ans me include karlo

flower
flowr
flight
fl
Terminal
fl
'fl'



BABBA
BABBAR
BABY
BABA
BALA
stop
root
BaB
aB
B
root
B
A
B
L
Y
B
A
A
A
R
temp = " "
"y"
tmp = " "
B
BA
BAR
tmp = " "
A
ans
y
BA
BAR
A
prefix → BAB
Y
BA
BAR
A
BABY
BABBA
BABBAR
BABA


flow
flower
f
l
o
w
e
r
i


ba
bala
baba
babu
Baby

Recursion

DP

3-4 lines

1 kal

DP

2 weeks

Wednesday

Part